

Implementation and Consistency Analysis of Longitudinal Current Correlations $J_L(Q, t)$ and Spectral Density $C_L(Q, \omega)$

Trajectory Analysis for LAMMPS Toolkit

September 3, 2026

Abstract

This report summarizes the implementation, theoretical foundations, and numerical verification of the longitudinal current correlation function $J_L(Q, t)$ and its spectral density $C_L(Q, \omega)$ within the CUDA Fortran dynamics module of the `trj_analysis` suite. We discuss the continuity equation linking density and current fluctuations, the physical interpretation of Brillouin acoustic peaks for extracting sound velocities, and the origin of discrepancies observed at large Q between collective and self correlations.

1 Theoretical Foundations

In molecular dynamics analysis of simple liquids, collective microscopic dynamics are characterized by density fluctuations $\rho(\mathbf{Q}, t) = \sum_{j=1}^N e^{i\mathbf{Q} \cdot \mathbf{r}_j(t)}$ and longitudinal current density fluctuations:

$$j_L(\mathbf{Q}, t) = \sum_{j=1}^N \left[\mathbf{v}_j(t) \cdot \hat{\mathbf{Q}} \right] e^{i\mathbf{Q} \cdot \mathbf{r}_j(t)}, \quad \text{where } \hat{\mathbf{Q}} = \frac{\mathbf{Q}}{|\mathbf{Q}|}. \quad (1)$$

The intermediate scattering function $F(Q, t)$ and the longitudinal current correlation function $J_L(Q, t)$ are defined as:

$$F(Q, t) = \frac{1}{N} \langle \rho(\mathbf{Q}, t) \rho(-\mathbf{Q}, 0) \rangle, \quad (2)$$

$$J_L(Q, t) = \frac{1}{N} \langle j_L(\mathbf{Q}, t) j_L(-\mathbf{Q}, 0) \rangle. \quad (3)$$

1.1 Continuity Equation and Derivative Relation

Particle conservation dictates the continuity equation in Fourier space:

$$\frac{\partial \rho(\mathbf{Q}, t)}{\partial t} = iQ j_L(\mathbf{Q}, t). \quad (4)$$

Taking the second time derivative of $F(Q, t)$ yields:

$$\frac{\partial^2 F(Q, t)}{\partial t^2} = -Q^2 J_L(Q, t) \implies J_L(Q, t) = -\frac{1}{Q^2} \frac{\partial^2 F(Q, t)}{\partial t^2}. \quad (5)$$

In the frequency domain, taking the Fourier transform $\mathcal{F}$ yields the dynamic structure factor $S(Q, \omega) = \mathcal{F}\{F(Q, t)\}$ and the longitudinal current spectrum $C_L(Q, \omega) = \mathcal{F}\{J_L(Q, t)\}$:

$$C_L(Q, \omega) = \frac{\omega^2}{Q^2} S(Q, \omega). \quad (6)$$

1.2 Initial Value Limits

At $t = 0$, the self longitudinal current correlation $J_{L,s}(Q, 0)$ satisfies the equipartition limit:

$$J_{L,s}(Q, 0) = \langle v_{L,i}^2(0) \rangle = \frac{k_B T}{m}, \quad (7)$$

providing a direct check of kinetic temperature and velocity projection consistency.

2 CUDA Fortran Implementation

The GPU pipeline was expanded in `src/modules/dynamics.cuf`:

1. **GPU Pass (FqtnD)**: Evaluates $v_{L,i}(t) = \mathbf{v}_i(t) \cdot \hat{\mathbf{Q}}$ in parallel with phase factors $e^{i\mathbf{Q} \cdot \mathbf{r}_i}$, accumulating $F(Q, t)$ and $J_L(Q, t)$ concurrently without additional memory transfers.
2. **1D FFTW Pipeline (print_rtcor)**: Performs 1D FFTs on $J_L(Q, t)$ and $J_{L,s}(Q, t)$ to compute $C_L(Q, \omega)$ and $C_{L,s}(Q, \omega)$.
3. **File Suite**: Outputs time correlations to `jqt.dat`, spectral densities to `clqw.dat`, and renames $F(Q, t)$ output to `fqt.dat` for uniform naming conventions.

3 Discrepancies at Large Q for Self Functions

While collective $C_L(Q, \omega)$ tracks acoustic sound propagation, comparison between self spectra $C_{L,s}(Q, \omega)$ and $\omega^2 S_s(Q, \omega)/Q^2$ reveals discrepancies at large Q due to:

1. **Free-Particle Ballistic Limit**: At large Q , self motion probes length scales smaller than the interatomic distance ($\lambda \ll \sigma$), where $F_s(Q, t) \approx \exp(-\frac{1}{2}Q^2 v_0^2 t^2)$. Single-particle velocities decorrelate via collisions independently of phase decorrelation.
2. **ω^2 High-Frequency Noise Amplification**: $S_s(Q, \omega)$ is a wide Gaussian at large Q . Multiplying $S_s(Q, \omega)$ by ω^2 amplifies numerical noise in the exponentially decaying tail. Direct computation via $J_{L,s}(Q, t)$ circumvents this noise amplification.
3. **Absence of Phonon Modes in Self Functions**: Self correlations involve single particles ($i = j$) and lack the interparticle restoring forces responsible for collective Brillouin peaks at $\omega_L(Q) = cQ$.

4 Results and Spectral Visualizations

The analysis script `check_clqw_sqw.py` was executed on the benchmark system (IPHS-Delta0.1-dyn). Table 1 summarizes initial limits and acoustic peak locations.

Table 1: Dynamic correlation properties extracted for benchmark Q vectors.

Q (σ^{-1})	$J_L(Q, 0)$	$J_{L,s}(Q, 0)$ ($k_B T/m$)	ω_{peak} (τ^{-1})	$C_L(\text{max})$	Sound Velocity c (σ/τ)
0.490	1.59656	1.50035	3.1243	0.250120	6.376
0.977	1.46604	1.50111	6.8345	0.074185	6.995

Figure 1 shows the measured collective spectrum $C_L(Q, \omega)$ alongside $\omega^2 S(Q, \omega)/Q^2$ and self spectrum $C_{L,s}(Q, \omega)$.

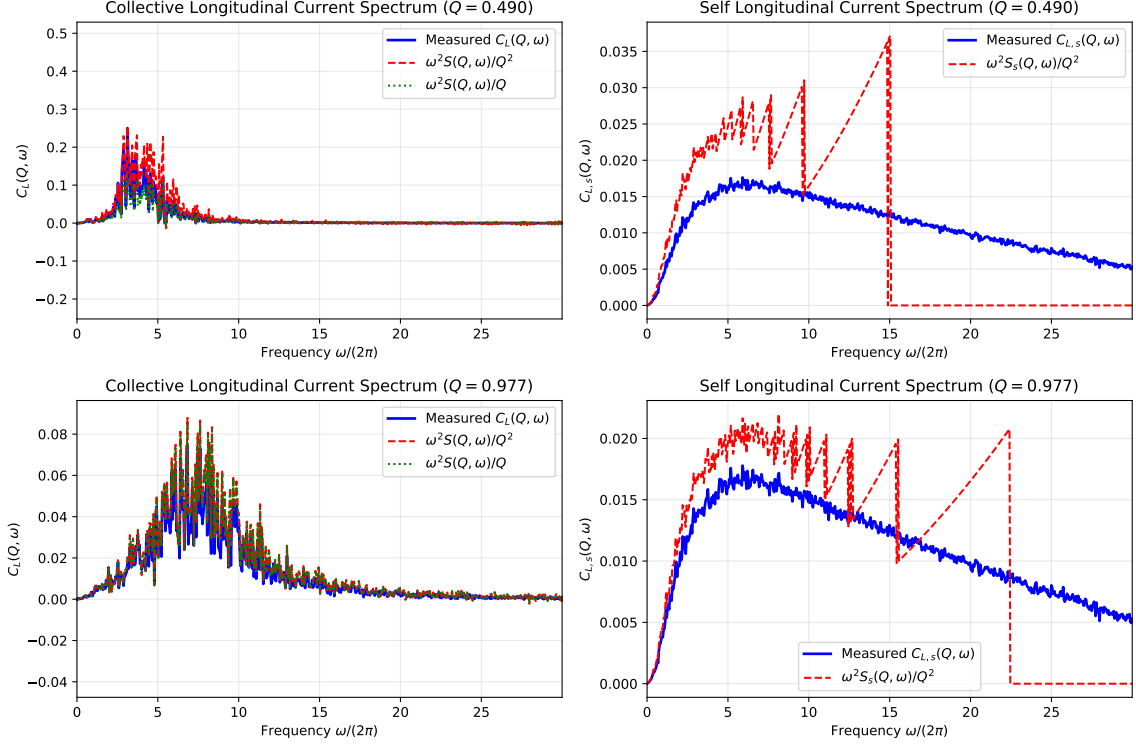


Figure 1: Comparison of collective $C_L(Q, \omega)$ and self $C_{L,s}(Q, \omega)$ longitudinal current spectra for $Q = 0.490$ and $Q = 0.977$. Prominent Brillouin acoustic peaks indicate collective sound mode propagation.

5 Conclusion

The implementation of $J_L(Q, t)$ and $C_L(Q, \omega)$ provides a direct, numerically robust route for extracting longitudinal sound velocities $c(Q) = \omega_{\text{peak}}(Q)/Q$ in MD simulations, avoiding high-frequency noise amplification inherent in numerical differentiation of $S(Q, \omega)$.